

JP's Portfolio

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1 Problem and Strategy Overview

Each team constructs and manages a maximum 10-stock equity portfolio drawn from the Nifty 100, Nifty Midcap 100 and Nifty Smallcap 100 universes, backtested from 1 January 2021 to 31 December 2025 with a virtual corpus of Rs 1 crore. The primary ranking metric is Total Net PnL, and a forward-looking out-of-sample stress test on 1 January 2026 to 30 June 2026 checks that the strategy generalises beyond the backtest window.

The central idea is a single, fully systematic composite-factor strategy applied identically to every eligible stock: a quarterly-rebalanced ranking on three price-based factors (12-1 month momentum, low-volatility, and a trend-quality measure), sized by inverse volatility with a concentration cap on the initial selection, with any leftover cash redeployed into the names already selected. The design was deliberately kept simple and fully rule-based, so it can be traced exactly how the portfolio was produced at any point in time. It was also validated using a train/test split and a rolling-window robustness check, rather than chosen purely for the highest in-sample return, in anticipation of the forward stress test.

2 Data

Universe. The eligible universe (≈ 300 stocks) was built by combining the current NSE constituent lists for Nifty 100, Nifty Midcap 100 and Nifty Smallcap 100. A handful of symbols appear on more than one list; these are de-duplicated to a single entry recording every index the stock belongs to. Since point-in-time historical index membership is not freely available, the current constituent lists are applied retroactively across the full 2021–2025 backtest; this is a stated simplification (see Limitations).

Prices. Daily OHLCV data for every universe stock and for the two benchmark indices was downloaded via the `yfinance` package, covering well before the 2021 backtest start through the end of the backtest. Prices are auto-adjusted closes (dividend- and split-adjusted). Only the adjusted *close* column feeds the model itself; open/high/low/volume are downloaded but not used by the current strategy. The extra pre-2021 history is a buffer so the 252-day momentum/quality lookback has sufficient data from day one of the backtest. At each quarterly rebalance, any stock with less than 95% price coverage over the lookback window (mostly recent IPOs) is excluded from that quarter's ranking rather than treated as having a zero or missing score.

Fundamental data. No fundamental data (earnings, book value, ROE, etc.) is used. This is a deliberate choice, not an oversight: reliable, point-in-time fundamental data free of look-ahead bias is not readily available for 300 stocks spanning large, mid and small caps in India. Building the model on price data alone also keeps the same formula applicable to every stock without exception.

3 Methodology

3.1 Stock selection

Every quarter (first trading day of Jan/Apr/Jul/Oct), a composite score is computed for every eligible stock:

$$\text{Composite} = 0.5 \times [0.5 Z(\text{Momentum}) + 0.5 Z(\text{Low-Vol})] + 0.5 Z(\text{Quality})$$

where:

- **Momentum** is the 12-month price return skipping the most recent month (the standard “12-1” construction, which avoids short-term reversal effects).
- **Low-Vol** is the negative of the trailing 90-day daily return standard deviation.

- **Quality** is the R^2 of a linear trend fit to log-price over the same 12-month window, which penalises spiky, parabolic price paths in favour of smoothly-trending ones.

Each factor is cross-sectionally z-scored across all stocks with at least 95% price history over the lookback window; stocks without enough history (mostly recent IPOs) are automatically excluded that quarter. The 10 highest composite-score stocks are then selected. Section 3.3 works through this scoring for one real stock and rebalance date.

Why these three factors. Momentum and low-volatility are two of the most extensively documented factors in equity research, and they are deliberately complementary rather than redundant. Momentum drives the return by following stocks that are already trending, while low-volatility manages risk, since momentum alone is known to be prone to sharp reversals (a “momentum crash”) after a strong run. The quality/trend-smoothness factor was added for a specific, tested reason rather than assumed upfront. An earlier two-factor version of this strategy underperformed the benchmark in a blind 2025 test (Section 7), and the cause was concentration in momentum names with erratic rather than steady price paths—exactly the names most prone to reversal. Adding trend-quality was confirmed, not merely assumed, to fix that weakness (Section 7). Value and fundamental-quality factors were excluded for the data-reliability reason above. Size is already handled structurally by the universe spanning three cap segments. Liquidity was not added separately since it overlaps heavily with volatility. And a short-term reversal factor was rejected because it would work against momentum over a similar horizon.

3.2 Weighting

Selected stocks are sized inverse to their own trailing 90-day volatility (the same volatility used in the Low-Vol factor), so that size is set by each stock’s own risk rather than by how high it ranks on the noisier composite score. Raw inverse-volatility affinities are shifted down by their minimum (plus a small constant) before normalising, then capped at 15% of portfolio value per name. Any excess above the cap is redistributed proportionally among the remaining names, iterated until no name exceeds the cap. This keeps a single very low-volatility name from dominating the book. Section 3.3 continues the worked example with this stock’s weight assignment.

The 15% cap governs this initial selection and sizing pass only. Because the ten target weights sum to exactly 100% while every buy also carries the 0.1% transaction cost on top, fully funding all ten names would need slightly more than the available capital; the lowest-priority name is typically left short once whole-share flooring is applied (Section 7). Rather than let that shortfall sit idle as cash, it is redeployed into the names that were funded, proportional to their own target weights – a more concentrated bet on names already selected, rather than forcing capital into the unfunded (typically highest-volatility) name or leaving it unused. This redeployment pass is deliberately allowed to push a name above 15%: the cap is a selection-risk control, not a hard ceiling on what happens to leftover cash from an already-capped selection. This design was chosen over leaving the shortfall as cash and over a version of redeployment that re-enforces the 15% cap during top-up, after testing all three: it won on CAGR, Sharpe and the 2025 held-out test, and matched or led on the 20-quarter rolling hit-rate.

3.3 Worked example

At the 1 October 2025 rebalance, across that day’s ≈ 277 eligible stocks, 12-1 month momentum had cross-sectional mean -2.55% and standard deviation 26.06% . BHARTIARTL’s own momentum was $+7.45\%$, giving $Z(\text{Momentum}) = (7.45 - (-2.55))/26.06 = 0.38$. The same $Z = (x - \text{mean})/\text{s.d.}$ formula applied to its trailing 90-day volatility (Low-Vol -1.03% vs. a cross-sectional mean of -1.75% , s.d. 0.61%) gives $Z(\text{Low-Vol}) = 1.18$, and applied to its trend-quality R^2 (0.76 vs. a mean of 0.32 , s.d. 0.25) gives $Z(\text{Quality}) = 1.75$. Its composite score, $0.5 \times [0.5(0.38) + 0.5(1.18)] + 0.5(1.75) = 1.27$, ranked 9th overall, placing it in the top 10 selected that quarter (Table 2).

For weighting, BHARTIARTL’s trailing 90-day volatility was 1.03% , giving a raw inverse-volatility affinity of $1/0.0103 = 97.0$. Across the ten selected names that quarter, the lowest affinity (Force Motors) was

26.3; shifting every affinity down by this minimum plus a small constant ($\epsilon = 0.1$) gives BHARTIARTL a shifted affinity of $97.0 - 26.3 + 0.1 = 70.8$, which is $70.8/345.6 = 20.5\%$ of the shifted total across all ten names. Since this exceeds the 15% cap, BHARTIARTL is capped at exactly 15%, and the excess 5.5 percentage points is redistributed proportionally among the other nine names.

3.4 Rebalancing and risk management

Stock selection and target weights are decided quarterly (first trading day of Jan/Apr/Jul/Oct); positions are held unchanged between rebalances. Only the delta needed to reach the new target is traded for names that remain selected, while dropped names are fully sold and new entrants are bought to target. A selected stock already trading below its own 200-day average on the rebalance date itself is not bought that quarter – a one-time check at the point of purchase, not ongoing daily monitoring (Section 7). A flat 0.1% transaction cost is charged on the notional value of every executed buy and sell, deducted directly from cash, so it compounds into every subsequent decision rather than being a cosmetic adjustment applied at the end.

3.5 Consistency and parameter selection

The same formula, thresholds and rules are applied to every one of the ≈ 300 eligible stocks with no per-stock discretion. The one free parameter with any real tuning behind it—the 50% weight given to the quality factor—was selected using only 2021–2024 training data and full-backtest metrics, deliberately excluding the 2025 held-out test, via a 0.001-resolution grid search. That search confirmed the chosen weight sits inside a genuine stable plateau (0.499 and 0.500 are bit-identical on the full backtest; 0.501–0.503 sit within 0.1 percentage point of the same result) rather than an isolated, fragile spike. This distinction matters because it is easy to overfit a single free parameter to one backtest window.

3.6 Key assumptions

- Trades execute at the same-day closing price as the signal date (standard close-to-close convention).
- All trades are in whole shares (no fractional shares), with target rupee allocations floored to the nearest full share.
- Idle or exited cash earns 0%.
- Corporate actions are handled via Yahoo Finance’s auto-adjusted close series.
- No slippage, shorting or leverage is modelled beyond the stated 0.1% transaction cost.

4 Tools and Software Used

The entire pipeline is written in Python 3, using `pandas` and `numpy` for data handling and the day-by-day backtest simulation, and `yfinance` for price data acquisition. No machine-learning or numerical-optimisation library is used: the model is a transparent, rules-based factor score rather than a fitted or black-box one. This was a deliberate choice, given the requirement that another evaluator be able to understand exactly how the portfolio was produced. `matplotlib` was used for the chart in this report, and the report itself is typeset in L^AT_EX. The complete code is in the accompanying GitHub repository, https://github.com/Kumar20-21/Finesse_x_IITB-Citadel_Lab_Portfolio_Challenge (see the README there for how to reproduce every number in this report).

5 Results and Performance Metrics

For additional context beyond the required list, the strategy also shows Beta ≈ 1.05 and annualised Alpha of +32.96% against Nifty 500 (CAGR – Beta $\times$ benchmark CAGR; slightly below Table 3’s raw +33.80pp outperformance since Beta is above 1), an Information Ratio of 2.34, a Sortino Ratio of 3.38, a Calmar Ratio of 1.63, and $R^2 \approx 0.53$ against the benchmark.

Metric	Value
Final Portfolio Value	Rs 7.37 crore
Total Net PnL	Rs 6.37 crore
Absolute Return	637.19%
Annualised Return (CAGR)	49.15%
Maximum Drawdown	−30.07%
Sharpe Ratio (rf = 0%)	2.35
Gain-to-Loss Ratio	1.70
Accuracy (% profitable closed trades)	61.62%
Total Executed Orders	279 (141 buys, 138 sells)
Closed Round-Trip Trades	99
Average Trades per Stock	3.28
Turnover (cumulative, 5-year)	91.1× initial capital
Total Transaction Costs Paid	Rs 9.11 lakh

Table 1: Required Round 2 metrics, full 2021–2025 backtest.

Ticker	Company	Shares	Price (Rs)	Weight
NAVINFLUOR	Navin Fluorine International Ltd.	2,066	5,912.89	16.57%
BHARTIARTL	Bharti Airtel Ltd.	5,833	2,079.43	16.45%
EICHERMOT	Eicher Motors Ltd.	1,635	7,236.72	16.05%
FORTIS	Fortis Healthcare Ltd.	10,133	883.06	12.14%
MUTHOOTFIN	Muthoot Finance Ltd.	2,190	3,779.74	11.23%
LAURUSLABS	Laurus Labs Ltd.	7,316	1,106.90	10.99%
MANAPPURAM	Manappuram Finance Ltd.	22,353	306.66	9.30%
ASTERDM	Aster DM Quality Care Ltd.	8,652	614.16	7.21%
FORCEMOT	Force Motors Ltd.	1	20,566.00	0.03%
Cash	(uninvested)	N/A	N/A	0.04%

Table 2: Final portfolio composition and weights, as of 31 December 2025.

6 Benchmark Comparison

Benchmark choice. Nifty 500 (index `^CRSLDX`) is used as the primary benchmark because the eligible universe spans large, mid and small caps. A broad, ≈ 500 -name, all-cap index is a more representative comparator than a large-cap-only index such as Nifty 100. Nifty 100 is reported alongside for completeness.

	Strategy	Nifty 500	Nifty 100
CAGR	49.15%	15.36%	13.35%
Total Return, 2021–2025	637.19%	104.17%	87.07%
Maximum Drawdown	−30.07%	−18.84%	−17.56%
Sharpe Ratio (rf = 0%)	2.35	1.07	0.94

Table 3: Strategy vs. benchmark, full 2021–2025 window.

The strategy outperforms both benchmarks by a wide margin on CAGR and total return, at the cost of a somewhat deeper maximum drawdown than Nifty 500—expected for a concentrated 10-stock book relative to a ≈ 500 -name index, and discussed further below.

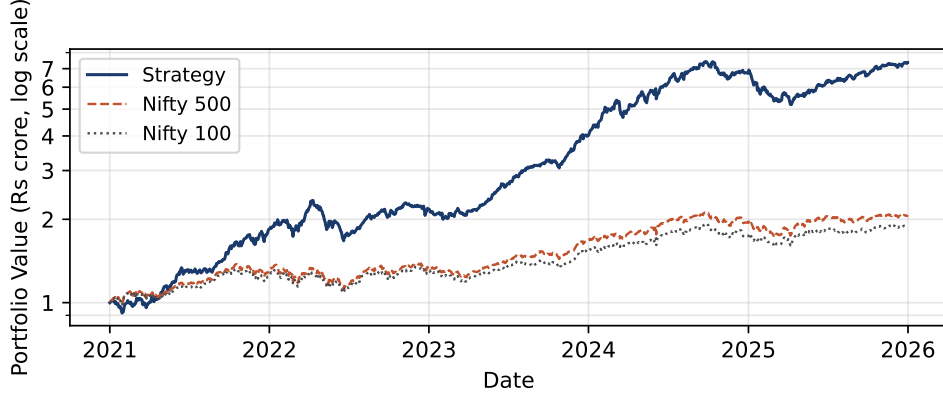


Figure 1: Portfolio value vs. Nifty 500 and Nifty 100, 2021–2025 (all series start at Rs 1 crore).

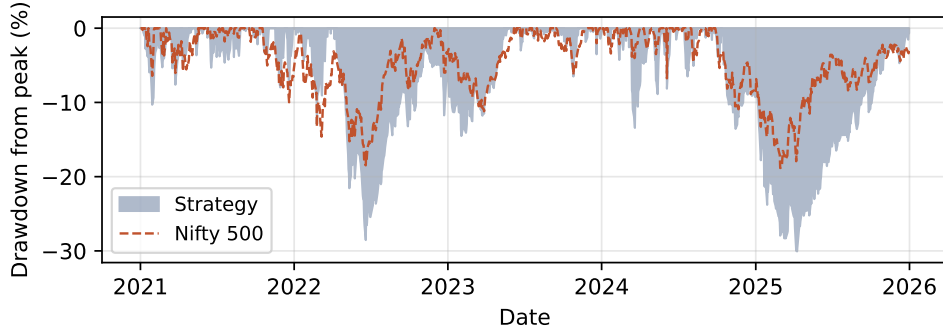


Figure 2: Drawdown from running peak, strategy vs. Nifty 500, 2021–2025. The strategy’s deepest trough is the -30.07% Maximum Drawdown reported in Table 1; it subsequently recovers to a new high.

7 Limitations and Discussion

Concentration and momentum-crash risk. Holding 10 stocks concentrates idiosyncratic risk far more than a 500-name benchmark, and a momentum tilt adds a specific failure mode on top: trending stocks can reverse sharply once a rally becomes crowded (a “momentum crash”). This is not theoretical – a two-factor version of this strategy (momentum and low-volatility only, no quality/trend-smoothness) returned -5.40% in a blind 2025 test, while Nifty 500 returned $+4.61\%$. Adding the quality factor fixed this in the same test ($+7.93\%$), but one held-out year does not guarantee immunity to a different kind of reversal in the future.

A tested and rejected risk control. A portfolio-level drawdown circuit breaker (de-risking 50% of the book whenever it fell more than 8% from its post-rebalance peak) was tested against a 20-quarter rolling grid. It shrank the worst single quarter’s excess over benchmark from -10.68pp to -9.00pp and reduced full-period maximum drawdown from -30.07% to -25.15% , and Sharpe actually improved slightly (2.35 to 2.50). However, it cut Total Net PnL from Rs 6.37cr to Rs 5.35cr and CAGR from 49.15% to 44.76%. Since Total Net PnL is the primary Round 2 ranking metric, this was a deliberate, metrics-driven decision to exclude the breaker, not an oversight. It reflects a real trade-off between shallower drawdowns and the metric the round is actually scored on.

Execution friction. All sells for a rebalance execute before any buys, so freed-up cash from trimmed or dropped positions is available to fund new and increased ones. Even so, because the ten target weights sum to exactly 100% of portfolio value while every buy also carries the 0.1% transaction cost on top, fully funding all ten names structurally needs slightly more capital than is available; the lowest-priority name is typically left short once whole-share flooring is applied. Rather than let that shortfall sit idle, it is redeployed into the names that were funded (Section 3.2). Selection also does not itself check trend status,

Year	Strategy	Nifty 500	Nifty 100
2021	+84.32%	+30.19%	+25.04%
2022	+18.31%	+3.02%	+3.63%
2023	+82.87%	+25.76%	+20.05%
2024	+68.81%	+15.16%	+11.76%
2025	+9.50%	+6.69%	+8.96%

Table 4: Calendar-year returns, strategy vs. benchmarks. The strategy leads in every year, including the weak-market year 2022, when both benchmarks were roughly flat.

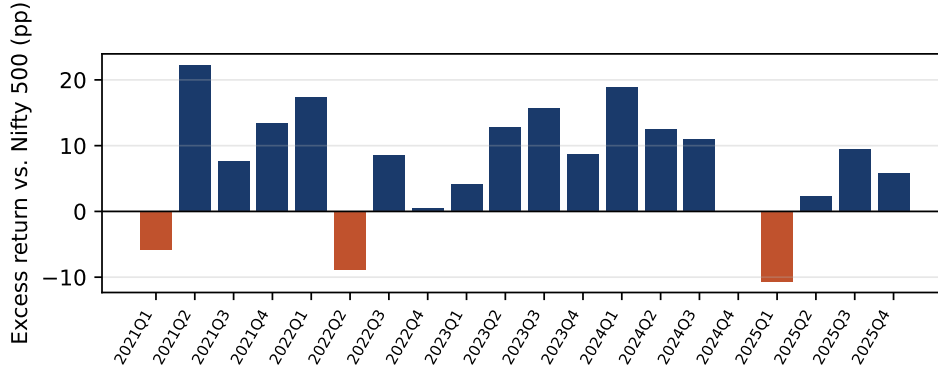


Figure 3: Excess return over Nifty 500 in each of the 20 independent calendar-quarter backtests used for the robustness grid above (the submitted strategy). 16 of 20 quarters (80%) beat the benchmark; the four losing quarters (2021Q1, 2022Q2, 2024Q4, 2025Q1) are visible as the bars below zero.

so a name can score well on momentum/low-volatility/quality while already trading below its own 200-day average on rebalance day; such a name is simply not bought that quarter (its allocation is zeroed), and may be selected again at a later rebalance once it recovers above its own average. Names being trimmed this quarter are also excluded from redeployment.

Universe timing. Using current (rather than historical, point-in-time) index constituent lists for the full 2021–2025 backtest is a simplification driven by data availability; a small number of names in the backtest may not have been constituents of their respective index for the entire period.

No fundamental data. The strategy is intentionally price-only. This avoids a real data-reliability problem for small and mid caps, but it also means the model cannot see valuation, earnings quality, or balance-sheet risk directly. It would not be expected to perform well in a regime where price-momentum and fundamentals diverge sharply.